Cooper Heyne Minehart

coopermineheart@gmail.com

651-387-3791

linkedin.com/in/cooper-minehart/

sites.google.com/view/cooperminehart/

Summary

Mechanical engineering PhD student at Carnegie Mellon University conducting ARPA-E supported research on technoeconomic analysis of battery manufacturing and cost projections under varying technology-commercialization and policy scenarios. Brings two years of university teaching experience across mechanical engineering curricula and industry experience in EV propulsion, instrumentation, and data-driven modeling.

Experience

Carnegie Mellon University – PhD Student
September 2025 - Current
Technoeconomic analysis of battery manufacturing. Projecting BEV energy storage costs under different tech-commercialization and policy scenarios. ARPA-E supported tech commercialization.

California Polytechnic University San Luis Obispo – Lecturer
September 2023 – September 2025
Full time lecturer in Mechanical Engineering. Courses taught below.

Empirical Systems Aerospace – Propulsion Team Manager
March 2022 – August 2023
Propulsion team manager for OEM flight test campaign. Expertise in EV instrumentation.

Mechatronic System Dynamics at KU Leuven - Researcher
May 2021 – March 2022 (10 months)
Flexible multibody contact models of wind-turbine gearboxes.

Researcher - Advanced Power Systems Laboratories
May 2019 – December 2020 (1 year 8 months)
Vehicle and dynamometer instrumentation, data-physics models for engine control. Two publications. Thesis: Data Driven Sensor Fusion for Cycle to Cycle IMEP Estimation

National Renewable Energy Laboratory - Vehicle Data Engineer Intern
May 2018 - Sep 2018 (5 months)
Data and physics models for emission estimates.

Inteva Products – Mechanical Design Engineer intern
May 2017 - Sep 2017 (5 months)

University of Minnesota – Data Acquisition Technician
May 2016 - Dec 2016 (8 months)

Education

Michigan Technological University, Master's — GPA: 3.78
Mechanical Engineering — Graduated Dec 2020
Master's degree with coursework in controls, digital signal processing, and dynamic systems. Research in machine learning applications for sensor fusion and data modeling.

Michigan Technological University, B.Sc. — GPA: 3.46
Mechanical Engineering — Graduated Dec 2019

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Skills

Signals, Simulation, and Controls
• Dynamic system simulation • Signal Processing (Python/MATLAB) • Flight Test • Controls/Simulation
CAD and Design
• NX (CAD, CAM, FEA, Modal Analysis) • Manual and CNC machining • 3D printing
Leadership
• Youth Mentoring • Project Management • Customer Relations

Courses Taught

Engineering Statics – Cal Poly San Luis Obispo
Analysis of forces on engineering structures in equilibrium. Properties of forces, moments, couples, and resultants. Equilibrium conditions, friction, centroids, area moments of inertia.

Philosophy of Design – Cal Poly San Luis Obispo
This course trains students to deconstruct and clearly define problems, ensuring that their proposals address every facet of the issue.

HM

Potential of Renewable Energy – Hochschule München
Climate change is an economic, engineering, social, and political problem. This course reviews each while focusing on the systems engineering issues of a Net Zero Energy future.

Engineering Design Communication – Cal Poly San Luis Obispo
Visual communication is the language of engineering. Students develop fluency in hand drafting and CAD through geometric principles, industry standards, and applied practice.

Introduction to Mechanical Engineering – Cal Poly San Luis Obispo
First year engineering students have a broad array of experiences and expectations. This course introduces students to the possibilities of a career in engineering.

Awards, Leadership, and Volunteering

AWIM Volunteer Teacher – Houghton Middle School — Oct 2017 - Dec 2019
Designed curriculum, organized volunteers, and taught a class of 8th grade students at HMS.
Reference: Sarah Geborkoff - Middle School Science Teacher - sgeborkoff@houghton.k12.mi.us

SAE Supermileage Engine Team Lead – MTU — May 2017 - Dec 2019
Leader of engine group at SAE Supermileage for Michigan Tech. Completed drive cycle optimization and composite-connecting-rod projects. Founded a CNC mentorship program.
Reference: Richard Berkey - Director of The Enterprise Program - rjberkey@mtu.edu

Certification Officer - Ridge Roamers Climbing Club — Sept 2017- Dec 2019
Trained members on safe technique and gear usage, hosted accessible climbing events, conducted belay-certification tests, and held open-climb.

Supermileage 1st Place in Design - SAE International — Jul 2018
Received first place in design at the 38th annual SAE Supermileage Competition.

Shell Eco Marathon Technical Innovation Award **Apr 2017**
Technical Innovation Award at the Shell Eco Marathon for work in clutch design on a BEV.

Publications

C. Heyne Minehart, J. R. Blough, J. D. Naber, C. Glugla, and C. Archer, “Sensor Fusion Approach for Dynamic Torque Estimation with Low Cost Sensors for Boosted 4-Cylinder Engine,” SAE World Congress, 2021.

C. Heyne Minehart, “Data Driven Sensor Fusion for Cycle to Cycle IMEP Estimation,” Michigan Technological University, 2020.